A Machine-Verified Zero-Parameter Temperature Scale Near Water's 4°C Density Anomaly: Theorem-Level Transition Scale and Hypothesis-Level Density Offset from the φ-Ladder

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Water's density maximum at $T_{\mathrm{obs}} = 277.15$ K (3.98 °C) is a central open problem in condensed matter physics. We derive, within the Recognition Science (RS) framework, a unique zero-parameter cooperative H-bond transition scale

$$T_{\mathrm{c}} = \frac{\varphi^{-5}}{8\,k_{\mathrm{B}}\,\ln\varphi},$$

where $\varphi = (1+\sqrt{5})/2$ is forced by cost self-similarity, $E_{\mathrm{coh}} = \varphi^{-5}$ is fixed by the Fibonacci constraint on $D = 3$, and the factor 8 is the forced 8-tick recognition cycle. The theorem-level result, machine-verified in Lean 4, is the interval bound

$$T_{\mathrm{c}} \in (267,\, 275)\ \mathrm{K},$$

together with the uniqueness statement that the adjacent φ-ladder rungs lie below freezing and above boiling. We then introduce a clearly labeled hypothesis-level finite-width correction

$$T_{\mathrm{max}} = T_{\mathrm{c}}(1+\varphi^{-8}),$$

motivated by the octave width of the crossover, which yields $T_{\mathrm{max}} \in (272.6,\, 281)$ K and contains T_{obs}. Thus the present paper establishes a theorem-level zero-parameter transition scale near 4°C and a hypothesis-level mapping from that scale to the density maximum itself. The three dedicated Lean modules introduced for this result compile with zero `sorry` and zero new axioms.

INTRODUCTION

Water reaches its maximum density not at the freezing point but at $T_{\mathrm{obs}} = 277.15$ K (3.98 °C). This anomaly—crucial for aquatic ecosystems, geophysics, and industrial processes—remains without a zero-parameter first-principles explanation [1, 2]. Phenomenological models (two-state [3], TIP4P/2005 [4]) reproduce the anomaly after calibration to experimental data, but they do not single out the relevant temperature by a zero-parameter structural argument.

In this Letter we make a more careful claim. Recognition Science (RS)—a cost-first framework whose algebraic core is published in *Axioms* [5]—fixes a unique zero-parameter *transition scale* near the water anomaly. That scale is theorem-level within the present RS bridge. Mapping that scale to the *density maximum itself* requires one additional finite-width hypothesis, which we state explicitly rather than fold into the theorem.

This separation is the main improvement over the previous draft: the manuscript now distinguishes theorem-level statements from hypothesis-level statements at every stage.

CLAIM STATUS

The present paper contains two logically distinct layers.

- **Theorem-level**: φ is forced by cost self-similarity, $D = 3$ and $2^D = 8$ are forced by the RS chain, $E_{\mathrm{coh}} = \varphi^{-5}$ is proved in `Foundation.CoherenceExponent`, and the formula $T_{\mathrm{c}} = \varphi^{-5}/(8k_{\mathrm{B}}\ln\varphi)$ together with the bounds $T_{\mathrm{c}} \in (267, 275)$ K and the adjacent-rung inequalities are proved in `Water.PhiLadderPartitionFunction` and `Water.DensityAnomaly`.
- **Hypothesis-level**: the finite-width correction $T_{\mathrm{max}} = T_{\mathrm{c}}(1+\varphi^{-8})$ and the inclusion $T_{\mathrm{obs}} \in (272.6, 281)$ K depend on a physical broadening ansatz rather than a derived RS equation of state.

The strongest theorem-level statement is therefore not "water is densest at exactly 277.15 K," but rather:

> *Within RS, there is a unique zero-parameter cooperative H-bond transition scale near the 4° C anomaly, and the next φ-ladder scales lie far outside the ambient liquid-water regime.*

We then study one specific, clearly labeled hypothesis that maps this transition scale to the density maximum itself.

THE FORCING CHAIN

The derivation uses five ingredients, each forced by the RS chain T5–T8. We state them in logical order; full proofs are in the Lean modules cited.

The golden ratio φ (T5–T6)

The Recognition Composition Law

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y) \tag{1}$$

together with normalization $J(1) = 0$ and calibration $J''_{\log}(0) = 1$ uniquely forces [5]

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1. \tag{2}$$

Self-similarity on the discrete ledger then forces $x^2 = x + 1$, whose unique positive root is $\varphi = (1 + \sqrt{5})/2$.

Dimension $D = 3$ and the 8-tick period (T7–T8)

Non-trivial linking (Alexander duality) requires $D = 3$ [6]. The minimal ledger-compatible walk has period $2^D = 8$.

Coherence exponent: $E_{\rm coh} = \varphi^{-5}$ (Fibonacci constraint)

Both $D = 3$ and $2^D = 8$ are Fibonacci numbers ($F_4 = 3$, $F_6 = 8$). The Fibonacci recurrence gives the deficit

$$2^D - D \;=\; F_6 - F_4 \;=\; F_5 \;=\; 5. \tag{3}$$

Independently, the J-cost integration measure has $D + 2 = 5$ independent variables (3 spatial + 1 temporal + 1 conservation). Both routes uniquely fix the coherence exponent:

$$E_{\rm coh} = \varphi^{-5} \;\approx\; 0.0902 \text{ eV}. \tag{4}$$

This falls squarely in the H-bond energy range (0.08–0.2) eV [11], a fact proved in Lean (`E_coh_in_water_hbond_range`).

Landauer cost per φ-bit

The RS Gibbs distribution $p(x) \propto \exp(-J(x)/T_R)$ motivates a Landauer-style information-energy bridge. Within the thermodynamic bridge formalized in Lean, the minimum cost to erase one φ-bit of ledger information at temperature T is taken to be

$$E_{\rm erase} = k_{\rm B} T \ln\varphi. \tag{5}$$

Here $\ln\varphi = J_{\rm bit} \approx 0.4812$ is the elementary ledger bit cost, forced by T5. The role of Landauer's principle is to connect this RS information scale to thermal energy [9, 10].

The theorem-level transition scale

The H-bond network maintains cooperative tetrahedral ordering as long as the total Landauer maintenance cost per 8-tick cycle remains below $E_{\rm coh}$. The cycle cost is

$$C_{\rm cycle}(T) = 8 \times k_{\rm B} T \ln\varphi. \tag{6}$$

Within the RS thermodynamic bridge, the cooperative transition scale is defined by the equality

$$C_{\rm cycle}(T_{\rm c}) = E_{\rm coh}. \tag{7}$$

This yields

$$\boxed{T_{\rm c} \;=\; \frac{\varphi^{-5}}{8\, k_{\rm B}\, \ln\varphi}} \tag{8}$$

with zero adjustable parameters. The Lean theorem proves that, once the bridge is accepted, the right-hand side is uniquely fixed.

NUMERICAL RESULT

Using the proved interval bounds $0.089 < \varphi^{-5} < 0.091$ and $0.481 < \ln\varphi < 0.483$ together with $k_{\rm B} = 8.6173 \times 10^{-5}$ eV/K, rigorous interval arithmetic gives the theorem-level bound

$$267 \text{ K} \;<\; T_{\rm c} \;<\; 275 \text{ K}. \tag{9}$$

The manuscript does not claim that Eq. (8) alone is already the exact density maximum. Instead, we consider a finite-width hypothesis motivated by the octave width of the crossover:

$$T_{\rm max} = T_{\rm c}\,(1 + \varphi^{-8}) \;\in\; (272.6,\, 281) \text{ K}, \tag{10}$$

with $\varphi^{-8} \approx 0.0213$, which places $T_{\rm obs} = 277.15$ K inside the interval. This correction is not theorem-level and should be read as a concrete, falsifiable proposal for how the structural transition broadens into the density maximum. The relative error of the uncorrected theorem-level scale is bounded: $|T_{\rm c} - T_{\rm obs}|/T_{\rm obs} < 4\%$.

UNIQUENESS OF THE AMBIENT-WATER LADDER SCALE

The φ-ladder is geometrically spaced: adjacent rungs differ by a factor of φ. We prove:

$$T_{\rm c} \times \varphi > 400 \text{ K} > T_{\rm boil}, \tag{11}$$

$$T_{\rm c}/\varphi < 184 \text{ K} < T_{\rm ice}. \tag{12}$$

TABLE I. Summary of the result, split into theorem-level and hypothesis-level layers.

Quantity	Result	Status	Observed
T_c (K)	(267, 275)	theorem	—
$\lvert T_c - T_{obs}\rvert / T_{obs}$	$< 4\%$	theorem	—
Adjacent rung up	> 400 K	theorem	steam
Adjacent rung down	< 184 K	theorem	solid ice
T_{max} (K)	(272.6, 281)	hypothesis	277.15
$\lvert T_{max} - T_{obs}\rvert$	< 5 K	hypothesis	—
Continuous fit parameters	0	both layers	—

No adjacent φ-ladder rung lies in the ambient liquid-water regime. Thus the RS ladder selects a unique *ambient-water scale*: one rung lower is deep in the solid regime and one rung higher is well into the vapor regime. This is the precise uniqueness statement proved in Lean.

PHYSICAL INTERPRETATION OF THE BRIDGE

The key physical insight connecting RS to condensed matter is what we call the *Landauer–φ bridge*. Standard Landauer theory [9] states that erasing one bit costs $\geq k_B T \ln 2$. In the RS ledger, the natural base is not 2 but φ; each rung of the φ-ladder encodes $\log_2 \varphi \approx 0.694$ classical bits. Hence the erasure cost per rung is $k_B T \ln \varphi$.

The H-bond network in water is proposed as a physical realization of the RS ledger: each water molecule participates in a tetrahedral coordination whose state is maintained by coherent H-bond oscillations. The 8-tick cycle provides 8 independent recognition slots, each requiring one φ-bit of maintenance. When $8k_B T \ln \varphi$ exceeds E_{coh}, thermal noise overwhelms cooperative ordering:

- *Below* T_c: Landauer cost $< E_{coh}$; the H-bond network sustains ice-like tetrahedral order (lower density).
- *Above* T_c: Landauer cost $> E_{coh}$; thermal fluctuations disrupt tetrahedral ordering, allowing closer packing.
- *Near* T_{obs}: a finite-width crossover can produce the density maximum once structural relaxation and packing are included.

The first two bullets are theorem-level within the bridge. The third is the physical interpretation that motivates Eq. (10).

FALSIFIABLE PREDICTIONS

The revised manuscript keeps only predictions that are tightly linked to the present derivation:

1. **A unique ambient-water crossover scale**: At ambient pressure there should be no second φ-ladder crossover between freezing and boiling. The nearest ladder scales are forced to lie below ∼184 K and above ∼400 K.

2. **Ultrafast H-bond dynamics near the anomaly**: The strongest crossover in network decorrelation should occur in the same temperature window as Eq. (9), with the slow molecular-gate timescale anchored near rung 19 (∼68 ps).

3. **Spectral concentration near 724 cm^{-1}**: The water librational mode near 724 cm^{-1} should be the most temperature-sensitive vibrational reporter of the crossover, because it is already fixed by the RS operating frequency.

4. **Tetrahedral templating changes the offset more than the base scale**: Perturbations that primarily affect packing geometry (confinement, templating surfaces, tetrahedral solutes) should shift the offset between T_c and T_{max} more strongly than they shift the base ladder scale T_c itself.

5. **No comparable anomaly without tetrahedral H-bond order**: Solvents lacking persistent four-fold H-bond coordination should not display an analogous crossover in the same RS manner, because the 8-slot cooperative maintenance mechanism is tied to tetrahedral network order.

MACHINE VERIFICATION

The entire derivation is formalized in three Lean 4 modules totaling ∼ 500 lines:

- `Foundation.CoherenceExponent`: Proves $E_{coh} = \varphi^{-5}$ from the Fibonacci constraint, with D=3 uniqueness verified by exhaustive check up to $D = 987$.

- `Water.PhiLadderPartitionFunction`: Proves the Landauer–φ transition condition (Eq. 8) and the ordered/disordered phase characterization.

- `Water.DensityAnomaly`: Proves the numerical bounds for T_c and T_{max}, the uniqueness inequalities (Eqs. 11–12), and packages the full certificate.

The three dedicated modules introduced for this paper compile with **zero** `sorry` (unproved obligations) and **zero new axioms**. The interval arithmetic uses only algebraic bounds on φ (via $\varphi^2 = \varphi + 1$) and Taylor-series bounds on $\ln \varphi$.

DISCUSSION

The strongest statement established here is not yet an exact theorem for the density maximum itself. What is established is a machine-verified zero-parameter structural transition scale near the anomaly, together with a precise uniqueness statement for the adjacent ladder scales.

The honest status of the derivation is as follows:

- **Theorem-level**: φ, the exponent 5, the factor 8, $\ln\varphi$ as the ledger information cost, the bridge formula $T_{\mathrm{c}} = \varphi^{-5}/(8k_{\mathrm{B}}\ln\varphi)$, the interval $(267, 275)$ K, and the adjacent-rung uniqueness statement.

- **Hypothesis-level**: the finite-width correction $T_{\max} = T_{\mathrm{c}}(1+\varphi^{-8})$ that maps the structural transition scale to the density maximum.

This distinction is not a weakness but a clarification. It turns the central experimental claim into a sharper target:

> *If the RS forcing chain and the Landauer–φ bridge are correct, then water must possess a unique zero-parameter cooperative H-bond transition scale near 4° C.*

The next step is clear: derive the finite-width mapping from the transition scale to the density maximum from an RS equation of state rather than introducing it phenomenologically. Whether RS is ultimately correct is an experimental question; the present paper isolates the part of that question that is already theorem-level and the part that remains open.

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[1] P. G. Debenedetti and F. H. Stillinger, *Supercooled liquids and the glass transition*, Nature **410**, 259–267 (2001).

[2] J. Russo and H. Tanaka, *Understanding water's anomalies with locally favoured structures*, Nature Commun. **5**, 3556 (2014).

[3] P. H. Poole, F. Sciortino, U. Essmann, and H. E. Stanley, *Phase behaviour of metastable water*, Nature **360**, 324–328 (1992).

[4] J. L. F. Abascal and C. Vega, *A general purpose model for the condensed phases of water: TIP4P/2005*, J. Chem. Phys. **123**, 234505 (2005).

[5] J. Washburn, *The Algebra of Reality: A Recognition Science Derivation of Physical Law*, Axioms **15**(2), 90 (2025).

[6] J. Washburn, M. Zlatanović, and E. Allahyarov, *Dimensional Rigidity: D=3 from Three Independent Proofs*, Recognition Science Technical Report (2026).

[7] J. Washburn, *The Water Computer Thesis: Liquid Water as Recognition Hardware*, Recognition Science Technical Report (2025).

[8] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Proc. CADE-28*, LNAI **12699** (Springer, 2021), pp. 625–635.

[9] R. Landauer, *Irreversibility and heat generation in the computing process*, IBM J. Res. Dev. **5**(3), 183–191 (1961).

[10] A. Bérut, A. Arakelyan, A. Petrosyan, S. Ciliberto, R. Dillenschneider, and E. Lutz, *Experimental verification of Landauer's principle linking information and thermodynamics*, Nature **483**, 187–189 (2012).

[11] J. D. Gaynor *et al.*, *Tracking hydrogen bond dynamics in water*, Nature Commun. **15**, 9753 (2024).